

DEVOPS SESSION THREE





PRESENTER INFORMATION

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DevOps Session 3

Session 3 Agenda

- Recap of foundational concepts
- Crafting a DevOps strategy
- Assessing organizational readiness
- Exploring adoption frameworks
- Understanding architecture basics

RECAP OF FOUNDATIONAL CONCEPTS

Week 2

Why DevOps Matters

The Three Ways

Essential Tools

Metrics Overview

Recap: Why DevOps Matters

A refresher on the core value of DevOps from Sessions 1-2.

- Unifies teams
- Accelerates delivery
- Reduces risks
- Enhances flexibility



Recap: The Three Ways

Reviewing the guiding principles from Session 1.

- Flow streamlines work
- Feedback improves quality
- Learning drives progress
- Culture supports success

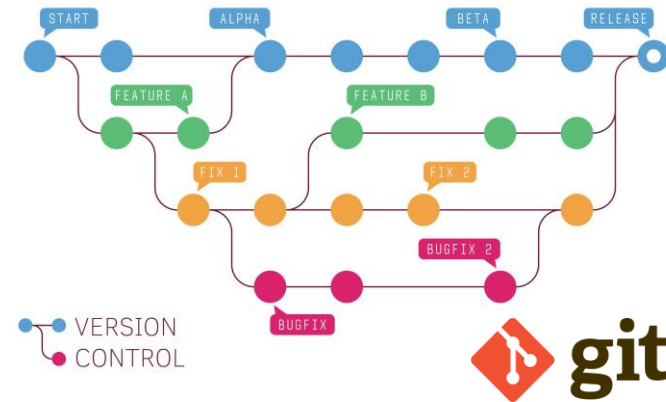


GENE KIM'S
THREE WAYS OF DEVOPS

Recap: Essential Tools

Key tools introduced in Session 2 for DevOps success.

- Git for version control
- Jenkins for automation
- AzureDevOps
- Docker for containers
- IaC for infrastructure



INFRASTRUCTURE AS CODE



STREAMLINED
PROCESSES



CHANGE
MANAGEMENT



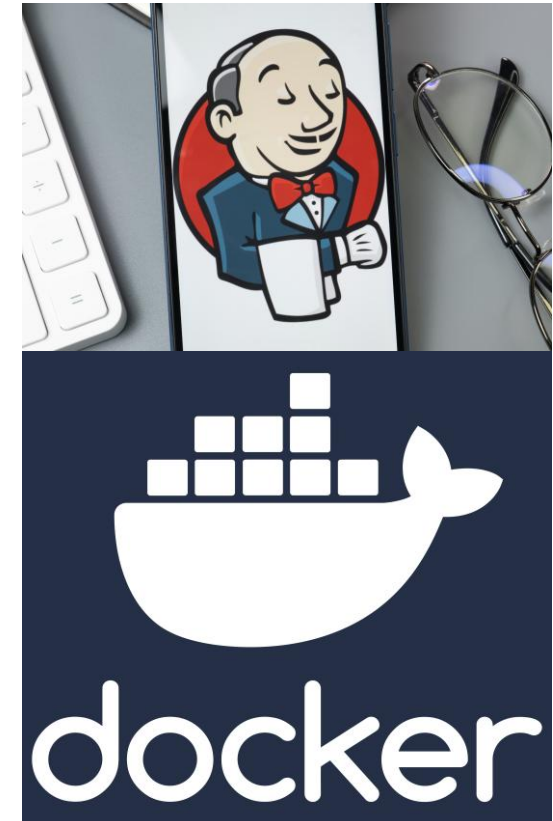
SCALABILITY



UNIFORMITY



CONFIGURATION



Recap: Metrics Overview

Core metrics from Session 2 to gauge performance.

- Lead time efficiency
- Deployment frequency
- Failure rate trends
- Recovery speed



FAILED



KPI

GROUP EXERCISE



Hit List group exercise.

- **Hit List Overview**
- **Hit List Steps 1-2**
- **Hit List Steps 3-4**

HIT LIST OVERVIEW

Hit List group exercise

- Identify inefficiencies
- Form small groups
- List key issues
- Set automation goals



HIT LIST STEPS 1-2



Start building the Hit List.

- **Brainstorm pain points**
- **Identify frequent tasks**
- **Note error-prone areas**
- **Draft initial list**

HIT LIST STEPS 3-4

Complete and present the Hit List.

- **Rank by impact**
- **Suggest automation fixes**
- **Prioritize top targets**
- **Share with the group**



❖ Strategy

- Why It's Essential
- Business Alignment
- SMART Goals
- Roadmap Steps
- Revenue Growth Example
- Customer Experience Example
- Avoiding Common Pitfalls



WHY IT'S ESSENTIAL

The critical role of a DevOps strategy.

Links to business goals

Prevents wasted effort

Sets clear priorities

Secures stakeholder support



BUSINESS ALIGNMENT



Connecting DevOps to organizational priorities.

- **Speeds market delivery**
- **Improves customer experience**
- **Drives cost efficiency**
- **Gains competitive edge**



SMART Goals

Setting clear and actionable DevOps objectives.

- Specific targets
- Measurable outcomes
- Achievable plans
- Time-bound deadlines



ROADMAP STEPS



Building a phased DevOps implementation plan.

Launch a pilot project

Scale successful efforts

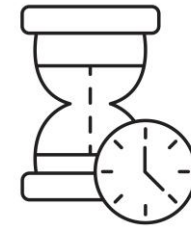
Gather ongoing feedback

Refine continuously

REVENUE GROWTH EXAMPLE

How DevOps enhances revenue opportunities.

- Rapid feature releases
- Reduced lead times
- Increased sales windows
- Retail case study



Processing Time



CUSTOMER EXPERIENCE EXAMPLE



Improving satisfaction with DevOps practices.

- Frequent system updates
- Consistent uptime
- High performance levels
- Service industry case

Avoiding Common Pitfalls

Mistakes to steer clear of in DevOps adoption.

- Overemphasis on tools
- Neglecting culture
- Skipping readiness checks
- Ignoring performance metrics



❖ Organizational Readiness

- Purpose of Assessment
- Evaluating Processes
- Cultural Assessment
- Technical Maturity
- Identifying Pain Points
- Assessment Tools



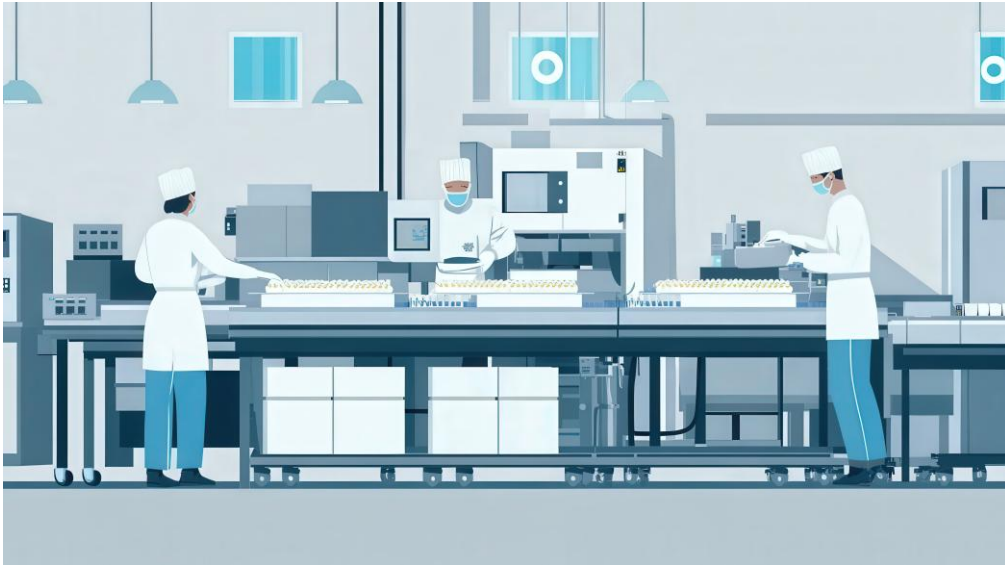
Purpose of Assessment

Why readiness evaluation is a critical first step.

- Identifies weaknesses
- Shapes strategic focus
- Conserves resources
- Builds stakeholder trust



Evaluating Processes



Analyzing current operational workflows.

- Map process flows
- Measure release cycles
- Identify bottlenecks
- Assess automation use

Cultural Assessment

Evaluating team mindset and preparedness.

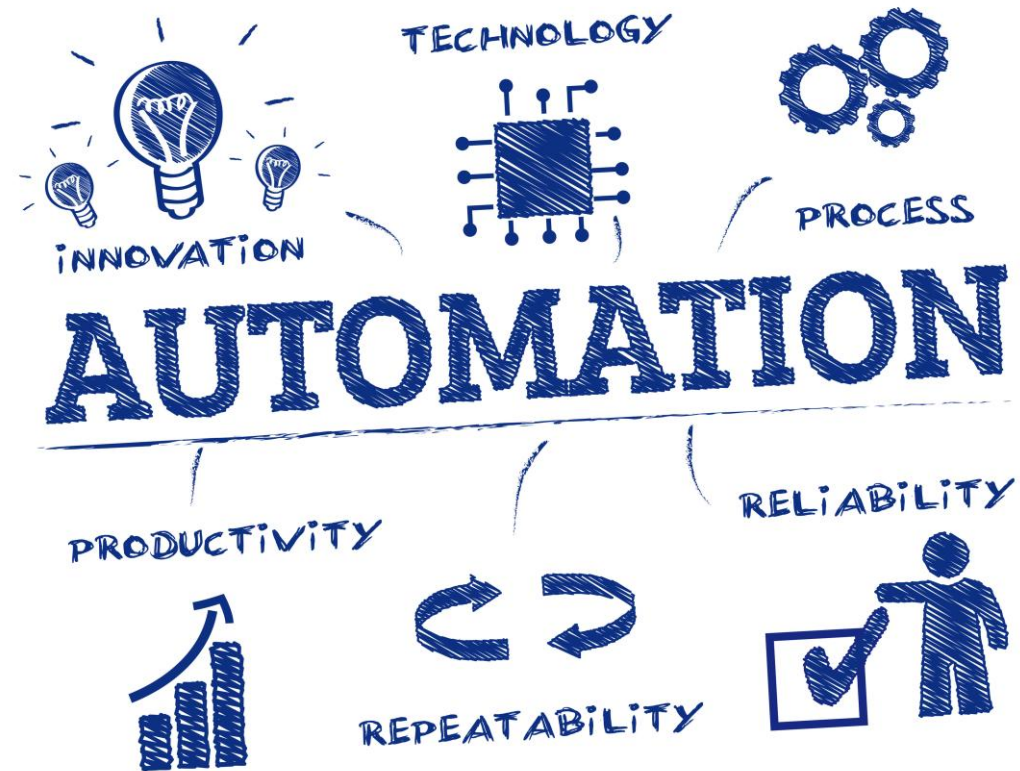
- Collaboration levels
- Response to failure
- Change adaptability
- Tools for cultural insight



Technical Maturity

Assessing current technical capabilities.

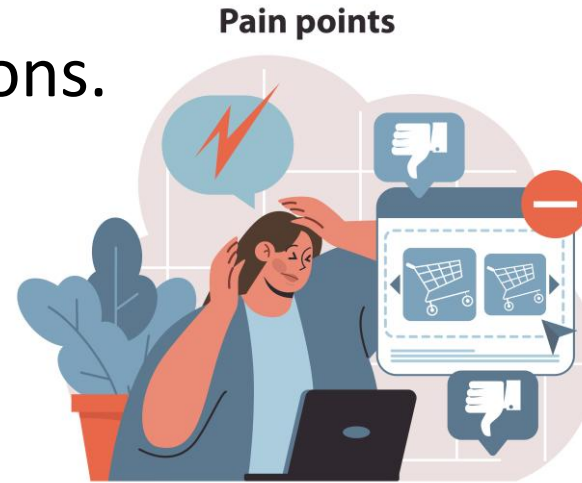
- Tool adoption status
- Automation coverage
- Skill proficiency levels
- System integration



Identifying Pain Points

Spotting inefficiencies in current operations.

- Manual repetitive tasks
- Frequent system outages
- Slow delivery cycles
- High error rates



4 Types of Customer Pain Points



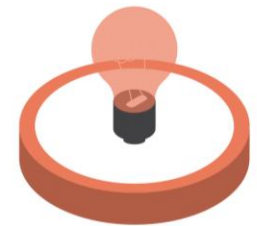
Process



Financial



Support



Productivity

Assessment Tools

Frameworks and tools to measure readiness.

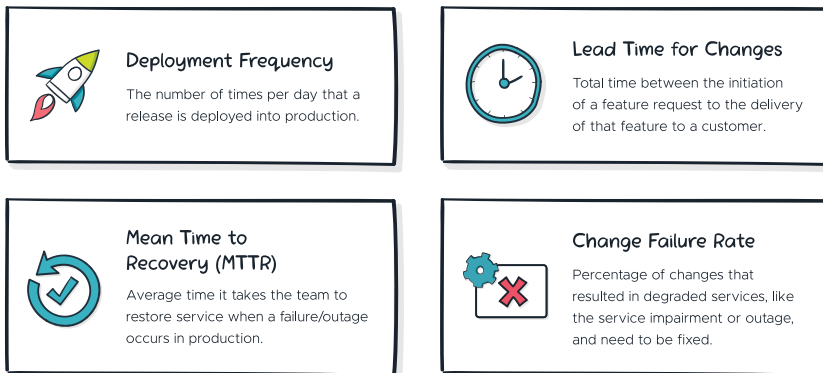
- CALMS framework
- DORA metrics analysis
- Feedback surveys
- Process audits



❖ FRAMEWORKS



Key DORA Metrics



- Purpose and Value
- CALMS Overview
- CALMS Sharing
- DORA Metrics Intro
- DORA Benchmarking

Purpose and Value

Why adoption frameworks are essential for success.

- Provide structured guidance
- Prioritize key focus areas
- Enable progress tracking
- Foster team alignment



CALMS Overview

Introduction to the CALMS (Culture, Automation, Lean, Measurement) model components.

- Culture emphasis
- Automation priority
- Lean efficiency focus
- Measurement standards



CALMS Sharing

The Sharing pillar of the CALMS framework.

- Knowledge dissemination
- Lessons learned exchange
- Cross-team alignment
- Collaboration tools



DORA Metrics Intro

Introduction to the DORA (DevOps Research and Assessment) metrics framework.

- Lead time efficiency
- Deployment frequency
- Mean time to recovery
- Change failure rate

Key DORA Metrics



Deployment Frequency

The number of times per day that a release is deployed into production.



Lead Time for Changes

Total time between the initiation of a feature request to the delivery of that feature to a customer.



Mean Time to Recovery (MTTR)

Average time it takes the team to restore service when a failure/outage occurs in production.



Change Failure Rate

Percentage of changes that resulted in degraded services, like the service impairment or outage, and need to be fixed.

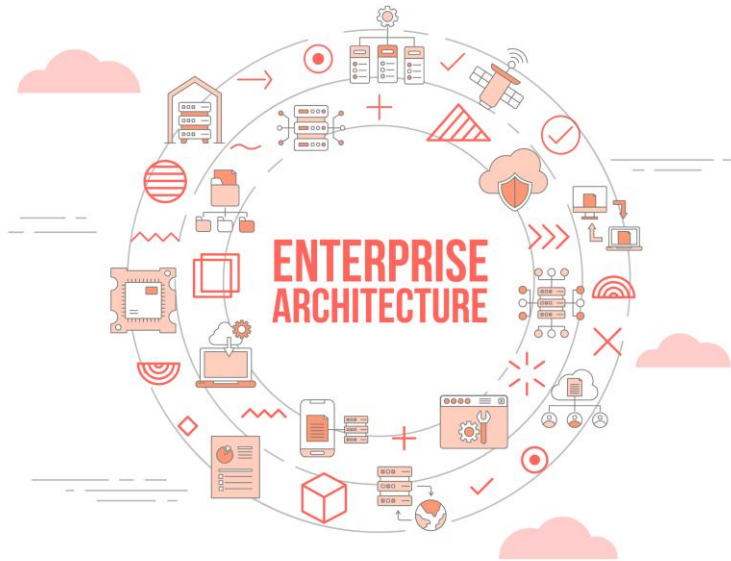
DORA Benchmarking

Using DORA to benchmark performance

- Compare to industry peers
- Set improvement targets
- Monitor trends over time
- Focus strategic efforts



❖ ARCHITECTURE



- What It Means
- Core Elements
- Pipeline Basics
- Integration Value

What It Means

Defining the role of DevOps architecture.

- Supports DevOps goals
- Enables rapid delivery
- Ensures scalability
- Maintains resilience

99.999%



Core Elements

Key components of DevOps architecture.

- Microservices design
- CI/CD pipelines
- Cloud infrastructure
- Monitoring systems

CI/CD PIPELINES



CONTINUOUS
INTEGRATION (CI)



CONTINUOUS
DEPLOYMENT (CD)



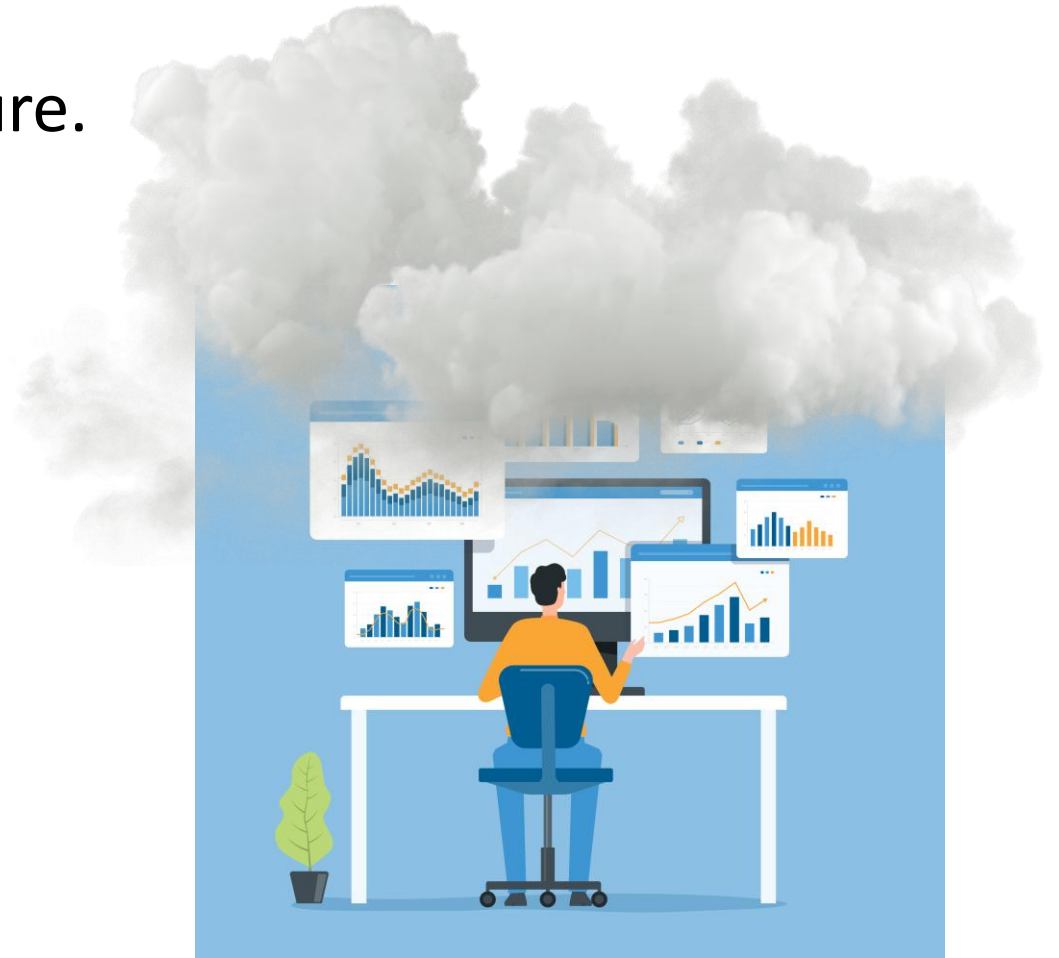
PIPELINE
FLOW



VERSION
CONTROL



AUTOMATED
TESTING



Pipeline Basics

The role of the delivery pipeline in DevOps.

- Automates builds
- Runs automated tests
- Deploys seamlessly
- Provides feedback loops



Integration Value

Why integration is critical in architecture.

- Synchronizes tools
- Ensures data flow
- Unites team efforts
- Reduces error rates



❖ Architectural Principles

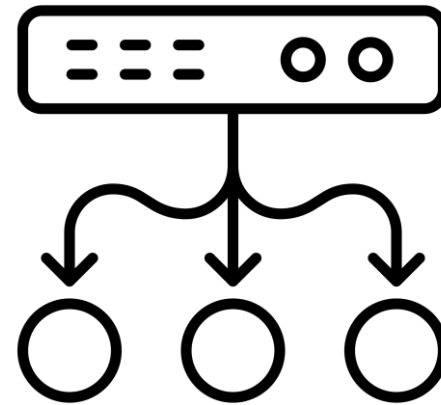
PRINCIPLES

- Scalability Design
- Reliability Practices
- Resilience Techniques

Scalability Design

Designing systems to handle growth effectively.

- Horizontal scaling
- Load balancing
- Elastic resource use
- Cloud utilization

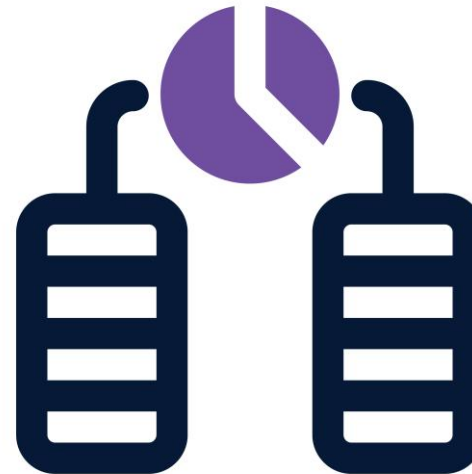


Reliability Practices

Ensuring consistent system performance

- Redundant systems
- Failover mechanisms
- Error management
- Uptime targets

99.999%



Resilience Techniques

Building systems that recover quickly.

- Failure planning
- Self-healing design
- Stress testing
- Rapid recovery focus



Demo

- Azure DevOps Pipeline Metrics Intro
- Azure DevOps Pipeline Metrics Features
- Azure DevOps Pipeline Metrics Insights



Azure DevOps Pipeline Metrics Intro

Introduction to the Azure DevOps Pipeline Metrics demo.

- Purpose of pipeline metrics
- Key metrics overview
- Dashboard setup preview
- Strategic insights goa

Key DORA Metrics



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Azure DevOps Pipeline Metrics Features

Key features showcased in the demo.

- Build success rate trends
- Deployment frequency graph
- Pipeline duration view
- Failed deployment alerts

Key DORA Metrics



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Quiz

Let's review what we discussed today



Question 1

What's a key benefit of a DevOps strategy?

- A) Increases manual effort
- B) Aligns with business goals
- C) Slows delivery cycles
- D) Reduces team collaboration

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Question 2

What does 'Lean' mean in CALMS?

- A) Speed focus
- B) Cost reduction only
- C) Efficiency emphasis
- D) Tool selection

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Question 3

Which DORA metric tracks recovery speed?

- A) Lead time
- B) Deployment frequency
- C) Mean time to recovery
- D) Change failure rate

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Question 4

What's a key scalability feature?

- A) Horizontal scaling
- B) Manual operations
- C) Fixed resources
- D) Single server reliance

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- **A) Horizontal scaling**
- B) Manual operations
- C) Fixed resources
- D) Single server reliance

Question 5

What's the Hit List exercise for?

- A) Tool inventory
- B) Spotting inefficiencies
- C) Budget planning
- D) Team assignments

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What's the Hit List exercise for?

A) Tool inventory

• **B) Spotting inefficiencies**

C) Budget planning

D) Team assignments